DESIGN AND IMPLEMENTATION OF CURRENCY RECOGNITION SYSTEM USING LABVIEW

S.Dhanya[#]1, Mrs.N.Kirthika[#2]

Department of ECE (PG- VLSI Design), Sri Ramakrishna Engineering College, Coimbatore, Tamilnadu, India

Abstract- There are many different countries all over the world having different currencies, with each of them looking totally different. To differentiate each of the currency we look at size of the paper used, color and pattern. The main aim of currency recognition system is used to classify the different currencies to help the people who need to recognize it, and work with convenience and efficiency. And also the classification of different currency is currently being used in many real time applications such as automated teller machine (ATM), chocolate vending machine and automatic ticketing systems. Moreover, as global economic integration proceeds, there is an increasing need for different currency classification. The system is based on personal computer with camera, which captures the image of currency and process them using pattern matching. Once the pattern is matched, using character recognition the equivalent INR value is displayed. The proposed system has to be implemented using LabVIEW.

Keywords-Pattern matching, LabVIEW.

I.INTRODUCTION

There are many different countries all over the world having many different currencies, with each of them looking totally different. For instance the size of the paper is different, the same as the color and pattern. The staffs who work for the money exchanging (e.g. Forex Bank) have to distinguish different types of currencies and that is not an easy job. They have to remember the symbol of each currency. This may cause some problems (e.g. wrong recognition), so they need an efficient and exact system to help their work. The aim of our system is to help people who need to recognize different currencies, and work with convenience and efficiency.

For bank staffs, there is a “Currency Sorting Machine”, which helps them to recognize different kinds of currencies. The main working processes of “Currency Sorting Machine” are image acquisition and recognitions. It is a technique named “optical, mechanical and electronic integration”, integrated with calculation, pattern recognition (high speed image processing), currency anti-fake technology, and lots of multidisciplinary techniques. It is accurate and highly-efficient. But for most staffs, they have to keep a lot of different characteristics and anti-fakes label for different commonly-used currencies in their mind. However, each of them has a handbook that about the characteristics and anti-fakes labels of some less commonly-used currencies. Even for that, no one can ever be 100 per cent confident about the manual

recognition. Otherwise currency recognition system is based on image processing using LABVIEW, which include extraction and pattern matching.

In order to make the system more comprehensive, we need to create a small database for storing the characteristics of the currency. In currency recognition system, we take Indian INR, Singapore SGD, Malaysian MYR and Sri lankan LKR as examples. The system can be programmed based on LabVIEW and it includes a user-friendly interface.

A literature survey shows that, fast and efficient algorithm to classify multi-national banknote images using size information and multi-template correlation matching [10], classification based on serial numbers in banknotes [2], currency classification based on texture analysis [8], to detect fake currency [1], etc.

The paper is organized as follows. Section II, explains the currency classification and identification method, Section III, deals with results and discussions and finally section IV concludes this paper.

II. CURRENCY RECOGNITION SYSTEM

We broadly classify our project into two modules as

- Currency classification
- Identification of denomination

A. Block diagram of the proposed work

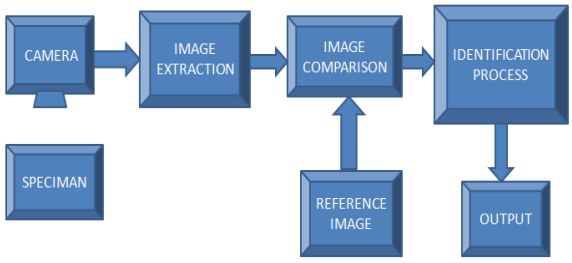


Figure.1. Block diagram of currency recognition system

In currency recognition system, the image of the currency is captured using camera as shown in figure 1 and the particular portion of the image is extracted from the input currency image. Before extracting the particular portion of the image, the reference image should be created. This can be done by VISION ASSISTANT. Then the extracted image from the captured currency image is compared with the reference image which is stored already as a template inside the LabVIEW. And also the equivalent INR value is displayed by identifying the special feature of the currency and using pattern matching for that image.

B. Currency classification based on pattern matching

First step is to capture the currency image using USB camera is done using IMAQdx snap2. After that the captured image is stored in IMAQ create function. For gray scale image processing, we can use IMAQ extract single color plane for extracting the luminance. Therefore we can compare different color currency which corresponding to same country. To check the image which has reference image or not, for we can create new template to compare it with processing image as shown in figure 2 to 5. We can use IMAQ match pattern to search a template image in the processing image. And also it displays the number of matches in the processing image. If the template is not matched then the

number of matches is displayed as zero. To store the result of pattern matching, we can use IVA store match pattern result.



Figure.2. Representing template of 100 INR



Figure.3. Representing template of 10 SGD



Figure.4. Representing template of 5 MYR



Figure.5. Representing template of 50 LKR

C. Identification of denomination value

To display the denomination value, we can identify the special feature of the currency. To check the image which has reference image or not, for we can create new template based on special feature to compare it with processing image. We can use IMAQ match pattern to search a template image in the processing image. And also it displays the number of matches in the processing image. If the template is not matched then the number of matches is displayed as zero. To store the result of pattern matching, we can use IVA store match pattern result. if the template of special feature is present in the processing image then it will displays the denomination value for particular currency. This will displays the denomination value as a string output.

Equivalent INR value for different country's currency:

- Singapore 1 SGD (dollar) = 46.44 INR*

- Srilanka 1 LKR (rupee) = 0.46 INR*

- Malaysia 1 MYR (ringgit) = 15.67 INR*

Special feature of 100 rupee of Indian currency from other Indian currency can be identifying by, the identification mark on left side of the 100 rupee is triangle as shown in figure 6.



Figure.6. Identification of 100 INR

Identification of 10 dollar of Singapore currency, the reverse side of $10 carries the theme of sports and

pictures of sportsman playing badminton, soccer, sailing, jogging and swimming as shown in figure 7. The equivalent INR value of Singapore dollar (SGD) is,

1SGD (Singapore dollar) = 46.44INR* (Indian rupee)

10SGD(Singapore dollar) = 464.4INR*(Indian rupee)



Figure.7. Identification of 10 SGD

Identification of 5 dollar of Malaysian currency, the left side of 5 dollar of Malaysian currency has the symbol which is different from other Malaysian currency of different denomination value as shown in figure 8. The equivalent INR value from Malaysian ringgit (MYR) is,

1MYR(Malaysian ringgit) = 15.67INR*(Indian rupee)

5MYR(Malaysian ringgit) = 78.35INR*(Indian rupee)



Figure.8. Identification of 5 MYR

Identification of 50 rupee of Sri lankan currency, the left side of 50 rupee of Sri lankan currency has the symbol which is different from other Sri lankan currency of different denomination value. For example, the 50 rupee of Sri lankan currency has 2 dots as an identification mark as shown in figure 9. The equivalent INR value from Srilankan rupee (LKR) is,

1 LKR (Sri lankan rupee) = 0.46 INR* (Indian rupee)

50 LKR (Sri lankan rupee) = 23 INR* (Indian rupee)



Figure.9. Identification of 50 LKR

III. RESULTS AND DISCUSSION

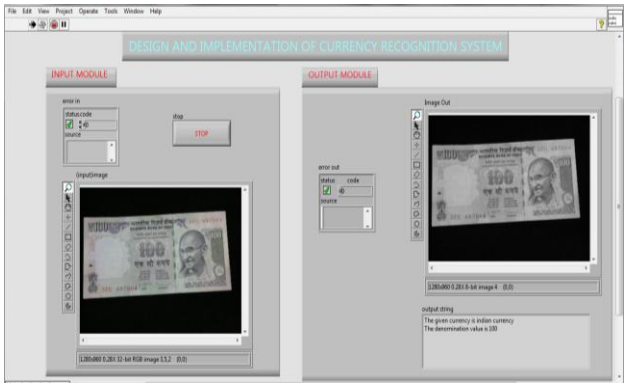


Figure.10. Simulation result of 100 INR

The Indian currency is given as an input image and corresponding country name and denomination value of 100 Rupee is displayed as shown in figure 10.

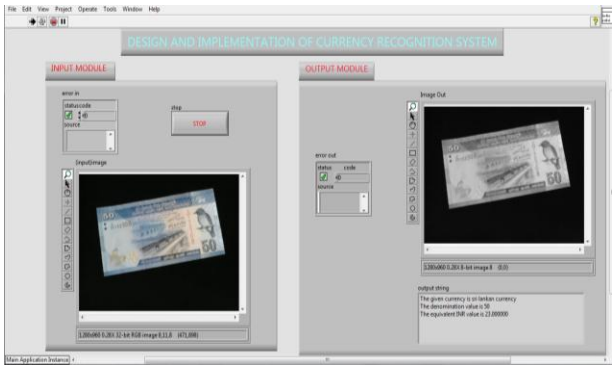


Figure.11. Simulation result of 50 LKR

The Srilankan currency is given as an input image and corresponding country name and denomination value of 50 Rupee and equivalent INR value 23 Rupee is displayed as shown in figure 11.



Figure.12. Simulation result of 10 SGD

The Singapore currency is given as an input image and corresponding country name and denomination value of 10 dollar and equivalent INR value 464.4 Rupee is displayed as shown in figure 12.

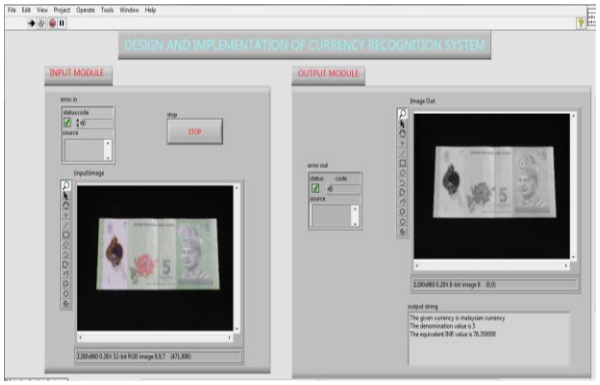


Figure.13. Simulation result of 5 MYR

The Malaysian currency is given as an input image and corresponding country name and denomination value of 5 Ringgit and equivalent INR value 78.35 Rupee is displayed as shown in figure 13. *The equivalent INR value for Singapore, Malaysia and Sri lanka were calculated as per the date on 09/08/2015.

IV. CONCLUSION AND FUTURE WORK

The currency recognition system to classify the different countries currency and to identify the denomination value based on special features and to display its denomination value equivalent to INR value is designed using LabVIEW 2014. The testing is done on four countries currency, taken as samples using USB camera. The captured input currency image is processed based on pattern matching and then the resultant output image can be displayed. With this the output string result also displayed which consists of corresponding country name and equivalent INR value of given input currency. From the result, it is clear that the designed system is working properly to classify the currency and to display the denomination value.

In the future, we are going to modify our system, overcome some limitation such as the light condition and complete our data base for recognizing more currencies. And also the currency recognition system implementation using MYRIO can be performed and also the number of samples to classify the currency and to display the equivalent denomination value can be increased.

V. REFERENCES

1. Binod Prasad Yadav, Patil.C.S, Karhe.R.R, Patil.P.H., " An Automatic recognition of fake Indian paper currency note using MATLAB", International Jouranl of Engineering Science and Innovative Technology(IJESIT),July 2014,pp. 560-566.

2. Bo-Yuan Feng, Mingwu Ren, Xu-Yao Zhang, Ching Y.Suen, “Automatioc recognition of serial numbers in bank notes”, International journal on pattern recognition,2014,pp. 2621-2634.

3. Chen.H.T., “Paper currency recognition system”, U.S. Patent 6639200, Kind Code B2, Oct 28, 2003.

4. Debnath.K.K., Ahmed.S.U., Shahjahan.M and Murase.K, “A Paper Currency Recognition System Using Negatively Correlated Neural Network Ensemble”, Journal of Multimedia, Vol. 5, No. 6, pp. 560-567, Dec. 2010.

5. Farid Garcia-Lamont,Jair Cervantes,Asdruba Lopez, “Recognition of Mexican banknotes via their colour and texture features”, International Journal on Expert Systems with applications,2012,ppt.9651-9660.

6. Frosini.A , Gori.M and Priami.P , “A Neural Network- Based Model for Paper Currency Recognition and Verification”, IEEE Transactions on Neural Networks, Vol. 7, No. 6, Nov. 1996.

7. Kishan Chakraborty, Jordan Basumatary, Debasmita Dasgupta, Jagadish Chandra Kalita, Subra Mukherjee, “Recent develpoments in paper currency recognition system”, International Journal of Research in engineering and Technology, 2013, pp. 222-226.

8. Kuldeep Verma, Bhupesh Kumar Singh, Anupam Agarwal, “Indian Currency Recognition based on Texture Analysis”, IEEE Transaction paper, 2011.

9. Reel.P.S , Krishan.G and Kotwal.S , “Image Processing based Heuristic Analysis for Enhanced Currency Recognition”, International Journal of Advancements in Technology, Vol. 2, No. 1, pp. 82-89, 2011;

10. Sungwook Youn, Euisun Choi, Yoonkil Baek, Chulhee Lee, “Efficient multi-currency classification of CIS banknotes”, International journal on Neurocomputing, 2015, pp. 22-32.